

Q1: 24 Feb (Shift 2) - Single Correct

The incorrect statement among the following is :

- (1) VOSO_4 is a reducing agent
- (2) Red colour of ruby is due to the presence of CO^{3+}
- (3) Cr_2O_3 is an amphoteric oxide
- (4) RuO_4 is an oxidizing agent

Q2: 24 Feb (Shift 2) - Single Correct

What is the correct order of the following elements with respect to their density?

- (1) $\text{Cr} < \text{Fe} < \text{Co} < \text{Cu} < \text{Zn}$
- (2) $\text{Cr} < \text{Zn} < \text{Co} < \text{Cu} < \text{Fe}$
- (3) $\text{Zn} < \text{Cu} < \text{Co} < \text{Fe} < \text{Cr}$
- (4) $\text{Zn} < \text{Cr} < \text{Fe} < \text{Co} < \text{Cu}$

Q3: 25 Feb (Shift 1) - Single Correct

In which of the following pairs, the outer most electronic configuration will be the same?

- (1) Fe^{2+} and Co^+
- (2) Cr^+ and Mn^{2+}
- (3) Ni^{2+} and Cu^+
- (4) V^{2+} and Cr^+

Q4: 26 Feb (Shift 1) - Single Correct

On treating a compound with warm dil. H_2SO_4 , gas X is evolved which turns $\text{K}_2\text{Cr}_2\text{O}_7$ paper acidified with dil. H_2SO_4 to a green compound Y. X and Y respectively are :

- (1) $X = \text{SO}_2, Y = \text{Cr}_2(\text{SO}_4)_3$
- (2) $X = \text{SO}_2, Y = \text{Cr}_2\text{O}_3$
- (3) $X = \text{SO}_3, Y = \text{Cr}_2\text{O}_3$

Questions with Answer Keys

MathonGo



Q5: 26 Feb (Shift 1) - Single Correct

Which one of the following lanthanoids does not form MO_2 ?

[M is lanthanoid metal]

(1) Nd

(2) Yb

(3) Dy

(4) Pr

Q6: 26 Feb (Shift 1) - Numerical

Dichromate ion is treated with base, the oxidation number of Cr in the product formed is :

Answer Key**Q1 (2)****Q2 (4)****Q3 (2)****Q4 (1)****Q5 (2)****Q6 (6)**

Hints and Solutions

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Q1 (2)

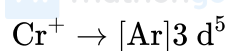
Red colour of ruby is due to presence of CrO_3 or Cr^{+6} not CO^{3+}

Q2 (4)

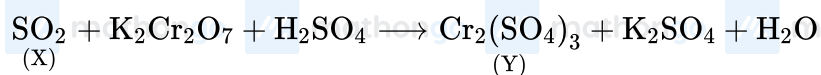
Fact Based

Density depend on many factor like atomic mass. atomic radius and packing efficiency.

Q3 (2)



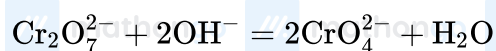
Q4 (1)



Q5 (2)

Fact

Q6 (6)



$$x + (-2 \times 4) = -2$$

$$x = 6$$